SYSTEM INDEX

B1 | Cable carrier + input buffers | Existing assembled modules remain external

PROJECT STRUCTURE

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03 02 MCU, MCP23017 and emergency stop

文件: SmartBin_B_02_01.kicad_sch

04 03 Serial module interfaces

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05 04 Buffered outputs and actuators

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08 06 Non-inverting sensor input translation

文件: SmartBin_B_06_02.kicad_sch

Assembly guidance and external module connections are specified in the accompanying manual.

Do not connect the core 3.3 V rail in parallel with the external 3.3 V regulator.

Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

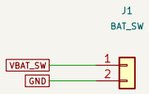
Sheet: /
File: SmartBin_B.kicad_sch

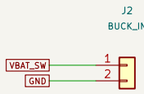
Title: SmartBin B / System index

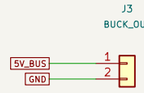
Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 1/8

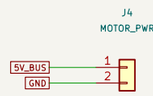
01 POWER DISTRIBUTION AND ADC

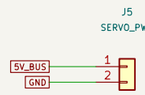
B1 | Cable carrier + input buffers | Existing assembled modules remain external

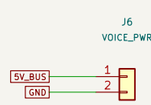




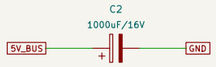


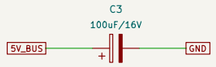














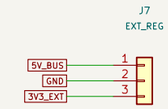


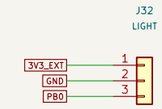


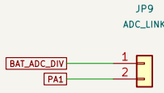












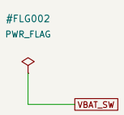


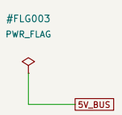


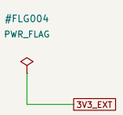


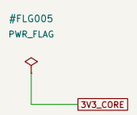












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.

STM32 + RFID / DIY through-hole carrier

Sheet: /02 01 Power distribution and ADC/
File: SmartBin_B_01_01.kicad_sch

Title: SmartBin B / 01 Power distribution and ADC

Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 2/8

02 MCU, MCP23017 AND EMERGENCY STOP

B1 | Cable carrier + input buffers | Existing assembled modules remain external









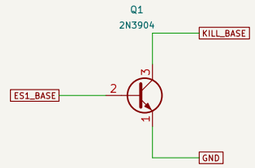


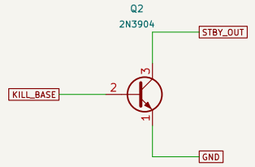


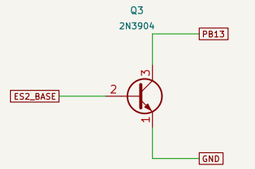


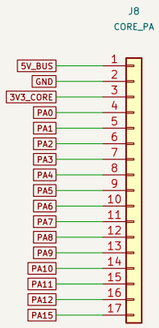


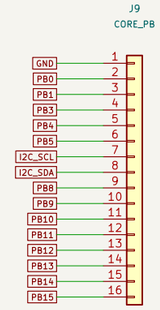


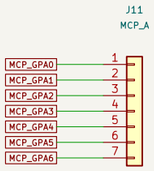


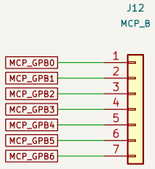


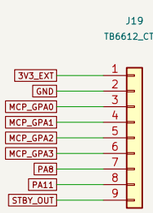


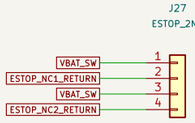












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /03 02 MCU, MCP23017 and emergency stop/
File: SmartBin_B_02_01.kicad_sch

Title: SmartBin B / 02 MCU, MCP23017 and emergency stop

Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 3/8

03 SERIAL MODULE INTERFACES

B1 | Cable carrier + input buffers | Existing assembled modules remain external







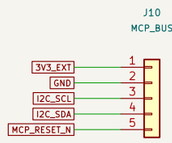


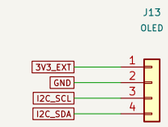


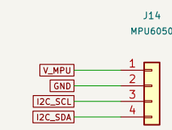


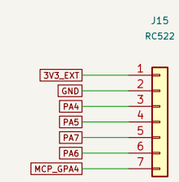


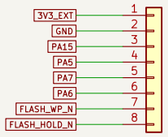


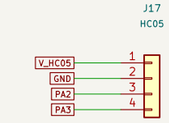


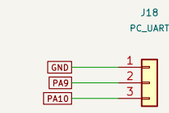


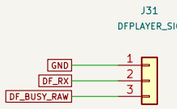


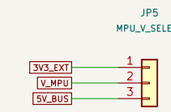


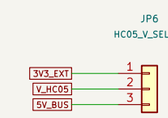


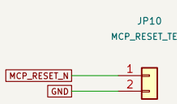












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /04 03 Serial module interfaces/
File: SmartBin_B_03_01.kicad_sch

Title: SmartBin B / 03 Serial module interfaces

Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 4/8

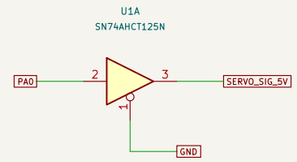
04 BUFFERED OUTPUTS AND ACTUATORS

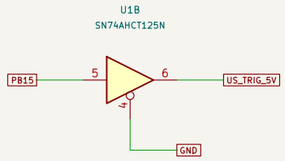
B1 | Cable carrier + input buffers | Existing assembled modules remain external

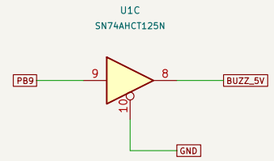


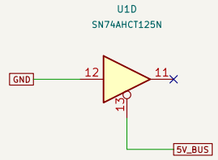


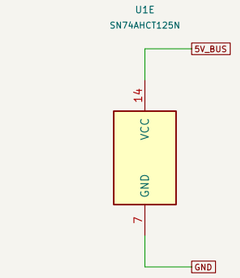
















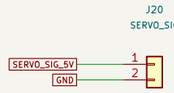


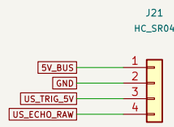


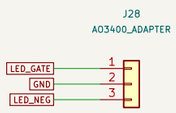




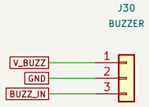


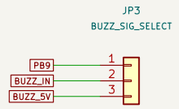


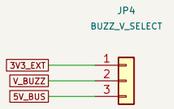












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /05 04 Buffered outputs and actuators/
File: SmartBin_B_04_01.kicad_sch

Title: SmartBin B / 04 Buffered outputs and actuators

Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 5/8

05 SENSOR AND USER INPUTS

B1 | Cable carrier + input buffers | Existing assembled modules remain external







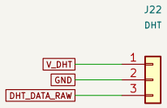


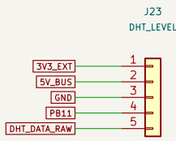


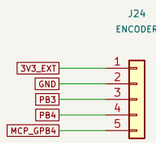


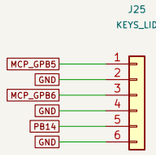


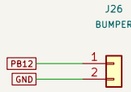


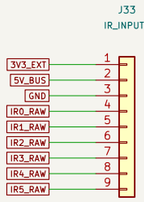


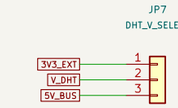


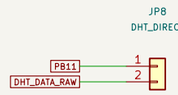












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

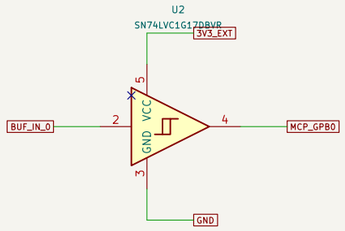
Sheet: /06 05 Sensor and user inputs/
File: SmartBin_B_05_01.kicad_sch

Title: SmartBin B / 05 Sensor and user inputs

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06 NON-INVERTING SENSOR INPUT TRANSLATION

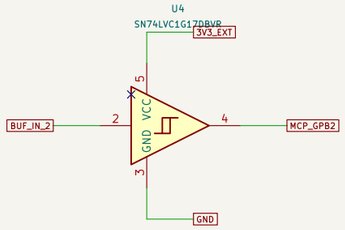
B1 | Cable carrier + input buffers | Existing assembled modules remain external







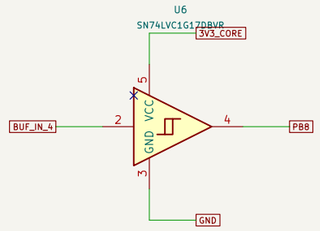








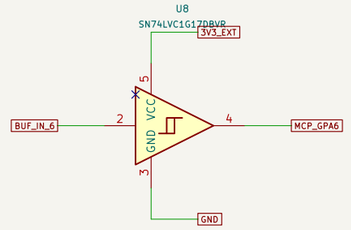


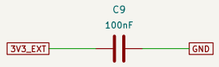






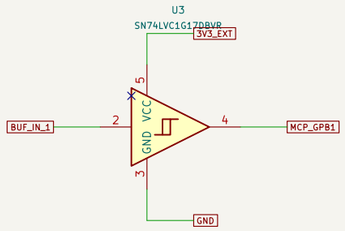








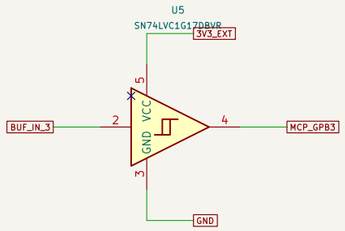


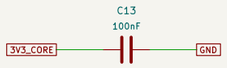






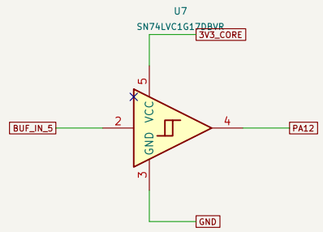








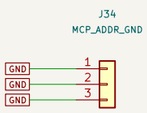












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.

STM32 + RFID / DIY through-hole carrier

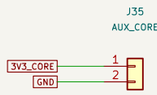
Sheet: /07 06 Non-inverting sensor input translation/
File: SmartBin_B_06_01.kicad_sch

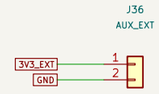
Title: SmartBin B / 06 Non-inverting sensor input translation

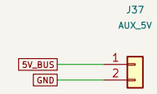
Size: A3	Date: 2026-11-03	**Rev: B1**
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06 NON-INVERTING SENSOR INPUT TRANSLATION

B1 | Cable carrier + input buffers | Existing assembled modules remain external







Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /08 06 Non-inverting sensor input translation/
File: SmartBin_B_06_02.kicad_sch

Title: SmartBin B / 06 Non-inverting sensor input translation

Size: A3	Date: 2026-11-03	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 8/8